DRIVER RISK PROFILING USING MACHINE LEARNING

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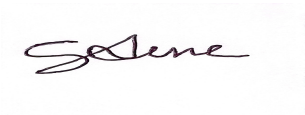
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Declaration

We do hereby declare that the research works presented in this thesis entitled, "DRIVER BEHAVIOR ANALYSIS AND RISK PROFILING USING MACHINE LEARNING" are the results of our works. We further declare that the report has been compiled and written by us, and no part of this report has been submitted elsewhere for the requirements of any degree, award, diploma, or any other purposes except for publications. The materials that are obtained from other sources are duly acknowledged in this thesis.

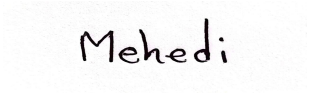
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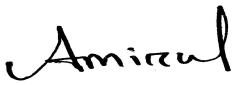
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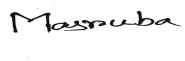
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Approval

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Dedication

We would like to dedicate this research to our loving parents . . .

Acknowledgement

We would like to express our heartfelt gratitude to the almighty Allah who offered our family and us kind care throughout this journey. We are deeply thankful to the Bangladesh University of Business and Technology (BUBT) for providing us with such a wonderful environment to pursue our research. We would like to express our sincere gratitude to Md. Shahiduzzaman, Assistant Professor, Department of CSE, BUBT. We have completed our research with his help. We found the research area, topic, and problem with his suggestions. He guided us with our study and supplied us with many research papers and academic resources in this area. He is patient and responsible. When we had questions and needed his help, he would always find time to meet and discuss with us no matter how busy he was. We also want to give thanks to Prof. Dr. Md. Ahsan Habib, Chairman, Department of CSE, and all our teachers for providing a solid background for our studies and research thereafter. They have been a great source of inspiration to us, and we thank them from the bottom of our hearts. We would also like to acknowledge our team members for supporting each other and being grateful to our university for providing this opportunity for us.

Abstract

Traffic collisions remain a pressing global safety problem and motivate the development of proactive, data driven interventions. Most conventional safety assessments rely on retrospective crash records or subjective observations and therefore provide limited opportunity for early intervention. This study develops a pipeline for driver behavior analysis that combines high resolution vehicle interaction traces from SUMO microscopic simulation with scenario variations generated by a Large Language Model (LLM). The dataset contains 49,136 time-step records from interactions involving more than 100 simulated vehicles. From these traces we derived kinematic features (notably Time-to-Collision and Time Headway) and combined them into a composite Risk Score. K-Means clustering produced four behavioral groups—Safe, Cautious, Normal, and Aggressive with a silhouette coefficient of 0.389, indicating a modest but useful separation. Three supervised classifiers Random Forest, XGBoost, and LSTM were trained on the cluster labels and evaluated using 5-fold cross-validation with SMOTE for class imbalance handling. All models achieved high accuracy: Random Forest attained the best performance at 98%, while XGBoost and LSTM each achieved 97%. SHAP analysis confirmed that the Risk Score and the rate of critical TTC events are the primary predictors. The framework therefore enables clear separation of high risk aggressive drivers from safer profiles, with potential applications in usage-based insurance, fleet safety programs, and autonomous vehicle decision making.

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List of Abbreviations and Acronyms

TTC Time-to-Collision

ML Machine Learning

RF Random Forest

SUMO Simulation of Urban MObility

SSM Surrogate Safety Measures

ADAS Advanced Driver Assistance Systems

ITS Intelligent Transportation Systems

UBI Usage-Based Insurance

TIT Time Integrated Time-to-collision

TET Time Exposed Time-to-collision

CSV Comma-Separated Values

IoT Internet of Things

AI Artificial Intelligence

GD Gap Distance

VF Follower Vehicle Speed

VL Leader Vehicle Speed

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Chapter 1

1 Introduction

Road traffic crashes remain one of the most pressing public-health problems worldwide. According to the World Health Organization, approximately 1.35 million people die in road traffic accidents every year, making it the eighth leading cause of death globally and the single largest killer of children and young adults aged 5 to 29 [1]. The economic toll is equally staggering: the National Highway Traffic Safety Administration places the annual cost of motor-vehicle crashes in the United States alone at over $340 billion when medical expenses, lost productivity, and property damage are combined [2]. In low- and middle-income countries, where the vast majority of fatalities occur, these costs can consume up to 3% of national GDP. Despite sustained engineering effort over several decades, progress in reducing these figures has been uneven.

For much of this history, automotive safety engineering focused on passive protection: structural crumple zones, seatbelts, and restraint systems such as airbags [3]. These technologies are designed to reduce occupant injury *during* a collision and have undoubtedly saved a large number of lives. However, they are inherently reactive, offering no benefit until an accident has already occurred. The subsequent introduction of active safety systems Anti-lock Braking Systems (ABS), Electronic Stability Control (ESC), and early forms of Advanced Driver-Assistance Systems (ADAS) marked a step toward collision avoidance, yet even these remain localized responses to individual kinematic events rather than holistic assessments of driving behavior [4].

A more fundamental shift has become possible with the widespread adoption of sensing and telematics technologies. Modern vehicles are equipped with GPS receivers, on-board diagnostics (OBD-II) ports, inertial measurement units, and camera- or radar-based ADAS. These systems continuously generate high-frequency

data describing vehicle interactions, offering insight into traffic dynamics at temporal resolutions that were previously unavailable. At the same time, smartphone-based telemetry leveraging built-in accelerometers and GPS at approximately 1 Hz—has emerged as a low-cost alternative for large-scale driving-data collection [5]. The result is an unprecedented volume of driving data: a rich opportunity for proactive safety strategies, but also a significant challenge in extracting meaningful risk information from noisy, high-dimensional streams.

This study builds on this data-rich context by examining driving behavior at a finer level of detail. Rather than relying on isolated event-based indicators, such as hard braking, it analyzes the continuous and often subtle interactions the so-called "dance" between neighboring vehicles. Machine learning techniques are employed to convert raw and noisy sensor data into interpretable measures of risk. The objective is to construct a behavioral fingerprint for each driver and position individuals along a spectrum ranging from "Safe" to "Aggressive." By identifying recurring patterns that tend to precede conflict, the proposed framework seeks to support interventions before a collision becomes imminent. Given that human error is implicated in more than 90% of road accidents [6], this work represents a step toward a more systematic and data-driven approach to reducing driver-related risk.

1.1 Problem Statement

The core challenge lies in effective measurement and interpretation. Although traditional approaches such as the analysis of historical crash records and observational studies provide valuable insights, they remain reactive and do not scale effectively [7]. At the same time, high-dimensional and high-frequency vehicle telemetry data are increasingly available, yet automated methods capable of translating this volume of information into a coherent representation of driving risk remain limited. A driver's average speed, for example, is an inadequate indicator of safety; consider a driver travelling at 120 km/h on a freeway with 200 m of headway versus one doing 100 km/h while sitting just 15 m behind the lead vehicle. The latter presents substantially greater collision risk despite the lower absolute speed. The most in-

formative indicators of risk are therefore embedded within the complex temporal interactions between vehicles, not in any single variable.

The scale of the data amplifies this difficulty. A single fleet of 1,000 vehicles, each equipped with sensors logging at 10 Hz, generates roughly 864 million records per day. Manual review is infeasible, and simple threshold-based alerts (e.g., "warn if TTC < 2 s") produce excessive false positives while ignoring the multi-dimensional character of risk. What is needed is an automated, interpretable pipeline that can process this volume and distil it into per-driver risk profiles capturing not just one but multiple concurrent unsafe behaviors.

Consequently, the central research question addressed by this study is: How can raw vehicle telemetry be automatically transformed into an interpretable, per-driver risk profile that integrates temporal proximity, spatial gaps, and acceleration behavior without reliance on lagging indicators such as historical crash involvement?

1.2 Problem Background

The challenge of modeling driver behavior resides at the confluence of physics, psychology, and data science [8]. From a physical perspective, vehicle motion is constrained by kinematic relationships; historically, researchers have sought to capture these constraints with physics-based car following formulations (for example, the Intelligent Driver Model [9]), which express following dynamics through ordinary differential equations. While these models are informative, they typically demand extensive calibration and often fail to represent the full range of human variability.

From a psychological perspective, driver actions reflect individual risk perception, transient emotional state, and ingrained habits [10]. Two drivers encountering the same traffic conditions may respond very differently depending on their experience, fatigue level, or cognitive distraction. In particular, the rise of smartphone-related distraction has introduced risk factors that traditional kinematic models do not capture. These psychological dimensions make driving an inherently stochastic process that deterministic equations alone struggle to describe.

Certain aggressive behaviors such as tailgating or abrupt acceleration are major

contributors to rear-end collisions, which remain the most frequent type of traffic crash. Metrics like Time-to-Collision (TTC) quantify immediate, objective hazard effectively [11], but they produce very large volumes of observations (millions of points per vehicle per day in high-resolution logs). Machine learning offers an advantage in handling this scale: unsupervised clustering can reveal emergent patterns of behavior by grouping drivers who exhibit similar temporal signatures, and supervised classifiers can subsequently be trained to recognize those patterns in new data. At the same time, the application of ML in safety-critical systems introduces a tension between predictive accuracy and interpretability a challenge this study explicitly addresses through the use of SHAP-based model explanations.

1.3 Research Objectives

Building on the gaps identified above the limitations of single-metric risk indicators, the need for automated processing of high volume telemetry, and the requirement for interpretable outputs—the primary aim of this study is to develop a machine learning pipeline that identifies, characterizes, and predicts driver behavior from high frequency vehicle telemetry. To achieve this aim, we pursue the following objectives:

1. **Engineer safety metrics.** Derive quantitative features from raw telemetry (notably Time-to-Collision (TTC) and Time Headway) and combine these physics-based indicators into a single, interpretable composite Risk Score that summarizes a driver's safety profile.

2. **Discover behavioral clusters.** Use K-Means clustering on the engineered feature set to reveal natural groupings in the data. This unsupervised stage is intended to produce distinct, interpretable profiles (for example, *Safe*, *Cautious*, *Normal*, and *Aggressive*) without relying on pre-labeled examples.

3. **Predict driver risk.** Train and validate multiple supervised classifiers Random Forest, XGBoost, and LSTM to assign new drivers to the cluster-derived profiles. Model evaluation will consider multiple metrics (accuracy, precision,

recall, and F1) and employ K-fold cross-validation with SMOTE to assess generalization and stability across different model architectures.

4. **Visualize risk boundaries.** Create clear visual representations (including crash envelope diagrams and multi axis radar plots) that expose the defining features of each profile and illustrate the thresholds separating safer from riskier driving behavior.

These objectives are addressed systematically in the subsequent chapters: feature engineering and algorithmic design are detailed in Chapter 3, and the experimental evaluation is presented in Chapter 4.

1.4 Motivations

The motivations for this study are both economic and technological. Road traffic collisions impose a substantial economic burden globally, with estimates by major public-health organizations placing the cost in the order of hundreds of billions of dollars per year. These losses encompass direct expenses (medical treatment, emergency response, and infrastructure repair) as well as indirect costs such as lost productivity and higher insurance expenditures. Such a large and persistent fiscal impact creates a strong incentive to adopt usage-based approaches to risk management. Usage-Based Insurance (UBI), enabled by in-vehicle telematics or smartphone telemetry, replaces static demographic proxies (e.g., age or address) with premiums tied to demonstrated driving behavior [12]. The behavioral analytics developed in this research supply the granular, interpretable measures required to support UBI schemes that reward safer driving and encourage risk-reducing actions.

From a technological standpoint, the transition toward mixed-traffic environments where human drivers and autonomous vehicles (AVs) coexist calls for systems that model and anticipate human behavior [13]. Safe AV operation depends not only on collision avoidance but also on the ability to infer intent: for example, recognizing an aggressive tailgater and adapting following distance, or anticipating a hesitant lane change from subtle motion cues. The driver profiles derived in this

work offer a practical foundation for such human-aware navigation strategies, which are critical to building public trust and enabling the safe integration of autonomy. In addition, these profiles have direct utility for fleet operators and safety managers: by identifying drivers with elevated risk signatures, organizations can target training and monitoring efforts to reduce accidents and operating costs.

1.5 Significance of the Research

This work impacts several parts of the transportation ecosystem:

i. **Traffic Safety:** It gives us a data-driven way to spot dangerous patterns before a crash happens [14].

ii. **Automotive Industry:** It helps build ADAS systems that can warn drivers based on their specific risk profile.

iii. **Insurance Sector:** It supports fair, transparent risk assessment for UBI.

iv. **Research Community:** It validates a methodology for processing car-following data, applying ML to safety-critical systems [15, 16].

1.6 Research Contribution

This study advances the field of driver behavior analysis by:

i. Developing a **Risk Score equation** that combines temporal proximity (TTC) and spatial gaps.

ii. Segmenting drivers into **four distinct profiles** using K-Means.

iii. Building a **multi-model comparison framework** with Random Forest, XGBoost, and LSTM classifiers, all achieving over 96% accuracy, with SHAP-based interpretability analysis.

iv. Implementing a robust **training strategy** using K-Fold cross-validation and SMOTE/class weighting to handle class imbalance.

v. Defining the **Crash Envelope** (Minimum TTC vs. Speed Difference) to visually map safe driving limits.

1.7 Complex Engineering Analysis

1.7.1 Complex Engineering Problems

This project confronted several engineering challenges in translating high-frequency, noisy telemetry into interpretable measures of human driving behavior. These challenges span signal processing, unsupervised inference, and design trade-offs required for both statistical validity and operational utility.

- **Defining Risk without Labels:** Unlike well-posed classification tasks where ground-truth categories are predetermined, "risk" is a behavioural continuum. Establishing a mathematical ground truth for aggressive versus safe driving required a physics-grounded clustering approach rather than pre-existing labels.
- **Data Dimensionality:** Raw trajectory data is noisy. Transforming it into signal-rich features like inverse TTC while managing outliers (like infinite values) required significant engineering.
- **Optimization Trade-offs:** Finding the right number of clusters meant balancing statistical validity (low variance) with practical utility.

1.7.2 Complex Engineering Activities

The research required the integration of multiple advanced engineering activities spanning data processing, algorithm design, and physical modeling:

- **Pipeline development.** A scalable data-processing pipeline was implemented to manage more than 49,000 vehicle-interaction frames, incorporating rule-based filters to remove noise and invalid observations.
- **Algorithmic tuning.** Clustering performance was evaluated across a range of cluster counts ($k = 2$ to 8), and Random Forest hyperparameters (including *n_estimators* and *max_depth*) were systematically tuned. Model behavior was

examined using confusion matrices to verify that predictions reflected meaningful risk-related patterns.

- **Mathematical modeling.** Kinematic relationships governing deceleration and headway were formulated and implemented, translating physical driving dynamics into computational features.

1.8 Thesis Organization

The rest of this thesis is structured as follows:

- **Chapter 2: Background Study** reviews the literature and theory.
- **Chapter 3: Research Methodology** details our framework, data processing, and algorithms.
- **Chapter 4: Implementation and Result Analysis** presents our experiments and findings.
- **Chapter 5: Standards, Constraints, and Milestones** discusses ethics, impact, and timelines.
- **Chapter 6: Conclusion** summarizes our work and suggests future directions.

1.9 Chapter Summary

This chapter has examined the need for automated approaches to driver behavior analysis in the context of increasingly data-rich traffic systems. It argued that prevailing safety assessment practices remain largely reactive and underutilize the high-resolution telemetry now routinely generated by modern vehicles. In response to this gap, the chapter segmented a framework centered on deriving a telemetry-based risk metric, using this measure to uncover distinct behavioral profiles, and enabling scalable risk assessment through supervised classification. The discussion also emphasized the engineering challenges associated with interpreting noisy, high-dimensional data and clarified the specific contributions of this work in addressing those challenges. Collectively, these considerations motivate the methodological

design choices developed in the following chapters and provide the basis for the experimental evaluation that follows.

Chapter 2

2 Background Study and Literature Review

This chapter situates the present research within the broader body of work on driver behavior analysis and traffic safety. It reviews key theoretical foundations and relevant prior studies, including the development of traffic safety metrics, traditional modeling approaches, and more recent machine-learning-based methods for driver profiling. The discussion progresses from general concepts toward increasingly specific techniques, ultimately highlighting the limitations of existing approaches and the research gap addressed by this thesis.

2.1 The Evolution of Traffic Safety Analysis

Research on traffic safety has changed significantly over time. Early studies focused primarily on analyzing historical accident records to understand why accidents occur. This approach is valuable for studying overall accident trends and injury patterns. However, it has an important limitation: accidents are rare events, and meaningful information is only available after an accident has already occurred.

As a result, only crash-based analysis provides limited support for real-time or preventive safety applications. To overcome this limitation, researchers have introduced Surrogate safety measures (SSMs), which focus on traffic collisions and near-miss situations rather than full accidents. The central idea behind SSMs is that risky interactions between vehicles occur much more frequently than accidents and can therefore be used to estimate collision risk in advance. One of the oldest and most widely used SSMs is the Time-to-Collision (TTC), which is defined as the time until a collision occurs if the vehicles continue at their current speeds and trajectories [17]. Smaller TTC values indicate more dangerous situations. Minderhoud and Bovy [14] later expanded this concept by introducing measures such as Time Exposed TTC (TET) and Time Integrated TTC (TIT), which measure how long

drivers are in a risky situation. Vogel [11] distinguished TTC from Time Headway, noting that headway reflects the potential for conflict, whereas TTC represents an actively evolving risk.

These concepts are consistent with Hydén's traffic conflict strategy [7], which demonstrated a clear relationship between near-miss events and actual crashes. Together, these measures form the basis of modern real-time safety analysis and inspire the risk metrics used in this thesis.

2.2 Methodological Approaches to Behavior Modeling

Modeling driver behavior requires an understanding of both how vehicles move and how people make decisions while driving. This problem draws on multiple disciplines, including dynamics, psychology, and statistical analysis. Over time, two broad methodological approaches have emerged in the literature. One work relies on **physics-based models**, which describe driving behavior through deterministic equations governing vehicle motion and interactions. These models provide important insights into the mechanical constraints of traffic flow but often rely on fixed assumptions about driver behavior.

The second line of research adopts **data-driven models**, which infer behavioral patterns directly from observed driving data using statistical and machine learning techniques. Rather than assigning explicit driving rules, these methods learn relationships from large-scale datasets and are therefore better suited to capturing variability across drivers and situations. As a result, data-driven methods have become increasingly prominent for modeling complex and uncertain aspects of human driving behavior.

2.2.1 Physics-Based Models

Early work on traffic behavior modeling focused heavily on car-following formulations, in which drivers are treated as deterministic agents responding according to predefined physical rules. Well-known examples include the Gipps model [18] and the Wiedemann model [19], both of which define vehicle acceleration and deceler-

ation as functions of relative speed and spacing to a leading vehicle. These formulations became foundational components of many microscopic traffic simulation tools.

Despite their usefulness for modeling vehicle motion, such approaches offer a simplified view of driving behavior. Real-world driving is shaped not only by kinematic constraints but also by psychological factors such as perception, attention, and individual risk tolerance. Because these elements are difficult to express through fixed equations, physics-based models often struggle to reflect the stochastic and diverse nature of human driving behavior [8].

2.2.2 Data-Driven Machine Learning Models

The increasing availability of vehicle sensor data has enabled a data-driven approach to driver behavior modeling. Rather than prescribing explicit driving rules, machine learning (ML) methods learn behavioral patterns directly from observed data, making them well suited to capturing subtle and individual driving characteristics. Higgs and Abbas [15] demonstrated that ML techniques can successfully segment and identify distinct driving patterns from trajectory data. Li et al. [20] further showed that these observable patterns are linked to drivers' underlying risk perception, supporting the view that driving styles are both consistent and measurable. More recently, Desai and Haque [5] illustrated that smartphone sensor data alone can be sufficient for meaningful driver profiling.

Deep learning has further extended the capabilities of data-driven driver behavior analysis. Dong et al. [21] applied recurrent and convolutional neural networks to characterize driving styles directly from raw time-series data, demonstrating that deep architectures can capture temporal dependencies that traditional feature-based models miss. Xing et al. [22] used LSTM networks for driver activity recognition in the context of ADAS, showing strong generalization across different driving scenarios. While these deep models achieve high predictive accuracy, their limited interpretability can be a disadvantage in safety-critical domains. This concern has driven growing interest in explainable AI (XAI) methods such as SHAP (SHapley

Additive exPlanations) [23], which provide both global and local explanations for model predictions. Parsa et al. [24] applied XGBoost with SHAP to highway safety analysis, demonstrating that interpretable feature attributions can coexist with high predictive performance. Consistent with this line of work, the present study adopts a data-driven methodology that combines multiple model architectures with SHAP-based interpretability to identify emergent behavioral patterns.

2.3 Key Enablers: Simulation and Data Generation

A persistent challenge in data driven traffic safety research is the availability of high quality data. Although naturalistic driving datasets provide valuable real world insight, they are costly to collect and often contain noise, missing values, and limited ground-truth information for safety critical events, which occur relatively infrequently. As a result, microscopic traffic simulation has become an important complementary tool for behavioral analysis.

The **Simulation of Urban Mobility (SUMO)** is a widely used, open source microscopic traffic simulator developed by the German Aerospace Center [25]. SUMO enables detailed modeling of complex traffic environments and supports the generation of large-scale vehicle trajectory data.

As noted by Krajewicz et al. [26] an important advantage of SUMO is its ability to generate precise and fully labeled trajectory information for each simulated vehicle. For driver behavior studies, this capability is particularly valuable, as it allows for controlled and repeatable vehicle-following situations and the extraction of clear dynamic variables such as speed, lag, and acceleration. These features make SUMO suitable for the development and validation of data-driven risk assessment models.

2.4 A Deep Dive into Driver Profiling Techniques

In data-driven driver behavior analysis, profiling is typically formulated as a two-stage learning problem. An unsupervised algorithm is first applied to discover characteristic driving styles present in the data. These styles are then used as labels for training a supervised model that classifies individual drivers into the corresponding

behavioral profiles.

2.4.1 Unsupervised Profiling via Clustering

In many driver behavior studies, explicit labels for categories such as "safe" or "aggressive" driving are not available, making unsupervised learning a natural starting point. Clustering methods aim to identify groups of drivers who exhibit similar behavioral characteristics without relying on predefined classes. Among these methods, **K-Means** remains one of the most widely adopted algorithms, originally introduced by MacQueen [27] and later formalized by Lloyd [28]. Its continued relevance in exploratory data analysis was highlighted by Jain [29].

Prior work commonly applies K-Means to partition drivers into a small number of clusters using kinematic features such as acceleration, jerk, and headway [30, 31]. The resulting clusters are then interpreted and assigned semantic labels (for example, "Timid," "Normal," or "Aggressive") based on the aggregate feature characteristics of each group.

2.4.2 Supervised Risk Classification

After behavioral clusters are identified and assigned semantic labels, they can be treated as reference classes for supervised learning. The objective is to train a model that maps the kinematic features of a previously unseen driver to one of the established behavioral profiles. Among commonly used classifiers, the Random Forest algorithm [32] is frequently adopted in driver behavior studies due to its strong performance on tabular data and its relatively transparent decision structure.

Although more complex models, such as deep learning approaches [33] or boosting methods like XGBoost [34], can achieve high predictive accuracy, their limited interpretability can be a disadvantage in safety-critical domains. In the context of Autonomous Vehicles (AVs), model transparency and verifiability are essential for safety validation [35] and for understanding system behavior [36, 37]. Frameworks such as Responsibility Sensitive Safety (RSS) [38, 39] further emphasize the need for formal and interpretable safety reasoning. Related behavioral prediction tasks

also play a key role in AV perception systems, which must anticipate the actions of surrounding human drivers [40].

2.5 Synthesis and Identified Research Gap

The reviewed literature highlights several consistent themes. Proactive safety assessment based on surrogate safety measures is widely recognized as essential, machine learning techniques have proven effective for analyzing driver behavior, and workflows that combine unsupervised clustering with supervised classification are commonly adopted. Table 1 summarizes the key characteristics of the studies reviewed above, enabling direct comparison of their data sources, methods, and scope.

Table 1: Summary of Related Work on Driver Behavior Analysis

Study	**Year**	**Data Source**	**Method**	**Key Limitation**
Higgs & Abbas	2013	Trajectory	Clustering	Single features
Constantinescu et al.	2010	OBD	Data Mining	No composite metric
Li et al.	2017	Survey+driving	Statistical	Small sample size
Desai & Haque	2021	Smartphone	ML	Limited features
Dong et al.	2016	Trajectory	CNN/RNN	No interpretability
Xing et al.	2019	Multi-modal	LSTM	No risk profiling
Ferreira et al.	2017	Kinematic	Multi-ML	Isolated variables
This study	**2026**	**SUMO sim.**	**K-Means+RF/XGB/LSTM**	**Simulated data**

Despite the progress summarized above, an important limitation remains. Many existing studies focus on a small number of isolated kinematic variables (for example, acceleration alone or headway alone), which does not fully reflect the combined nature of driving risk. High acceleration, for instance, may be benign on an empty roadway but becomes hazardous when it occurs alongside short following distances. Furthermore, the majority of reviewed studies either omit model interpretability entirely or treat it as an afterthought, which limits their applicability in safety-critical decision-making.

To the best of our knowledge, no existing study combines TTC rate, tailgating rate, and aggressive acceleration rate into a single weighted index for unsupervised driver profiling. This research is designed to address this gap directly. The central contribution is the formulation and validation of a multi-factor Risk Score that

integrates tailgating behavior, critical Time-to-Collision events, and aggressive acceleration into a single, interpretable metric. By using this composite score as the basis for clustering, the resulting driver profiles better capture the holistic nature of driving risk. The subsequent chapters present the methodology used to construct this metric and evaluate the performance of the proposed data-driven framework.

2.6 Chapter Summary

This chapter reviewed previous studies on driver behavior analysis and traffic safety. It showed how research has moved from crash-based analysis toward proactive safety measures. It examined both physics-based and data-driven modeling approaches and discussed the role of simulation and machine learning in enabling scalable behavior analysis. The review highlighted common driver profiling techniques based on clustering and supervised classification. Finally, a research gap was identified in the reliance on isolated kinematic variables, motivating the composite risk modeling framework developed in this study.

Chapter 3

3 Research Methodology

This chapter describes the methodology used to develop the driver profiling system. A structured pipeline is designed to transform high-frequency vehicle trajectory data generated by the SUMO simulator into meaningful behavioral profiles. The methodology covers the complete process, from simulation setup and data generation to feature extraction and predictive model training.

3.1 Proposed Framework

The proposed driver profiling framework is organized as a sequential pipeline that transforms simulated vehicle interactions into interpretable behavioral profiles. Each stage of the pipeline is designed to progressively refine the data and extract meaningful representations of driving behavior:

1. **Simulation configuration:** A road network and traffic demand were configured within the SUMO simulator. Traffic flow scenarios were generated using a Large Language Model (LLM) to introduce diversity in car following behavior and environmental conditions.

2. **Data generation:** The configured simulation was executed to produce high-frequency vehicle trajectory data. The logging frequency was selected to ensure that short term driver reactions and interaction dynamics were adequately captured.

3. **Data parsing and loading:** The simulator output, originally stored in XML format, was converted into structured CSV files. This transformation enabled efficient storage, querying, and processing using common data analysis tools.

4. **Preprocessing:** Several preprocessing steps were applied to improve data quality. These included removing records associated with vehicles entering or

exiting the network, filtering unrealistic kinematic values, and limiting extreme Time-to-Collision values to a meaningful interaction horizon.

5. **Feature engineering:** From the cleaned trajectory data, a set of physics based features was derived. These features include Time Headway, Required Deceleration, and other safety-related indicators, which were further combined to construct a composite Risk Score representing unsafe driving behavior.

6. **Driver profiling:** Time series observations were aggregated at the individual vehicle level to form statistical profiles. Each profile summarizes a driver's typical behavior across the engineered features, such as average following distance and frequency of critical interactions.

7. **Clustering:** The resulting driver profiles were grouped using the K-Means clustering algorithm to identify distinct driving styles. The number of clusters was selected empirically using inertia and silhouette metrics to balance separation quality and interpretability.

8. **Classification:** The cluster assignments were then treated as labels for supervised learning. A Random Forest classifier was trained to predict a driver's behavioral profile from their feature set and evaluated using a held-out test dataset.

9. **Visualization:** Finally, visualization techniques including crash-envelope plots were generated to support interpretation and comparison of the identified driver profiles.

3.2 Data Set Analysis

3.2.1 Justification for Simulation-Based Data

A persistent challenge in data-driven traffic safety research is the availability of high-quality labeled data. Naturalistic driving studies such as the 100-Car Study [41] provide invaluable real-world insight, but they are expensive to conduct (on

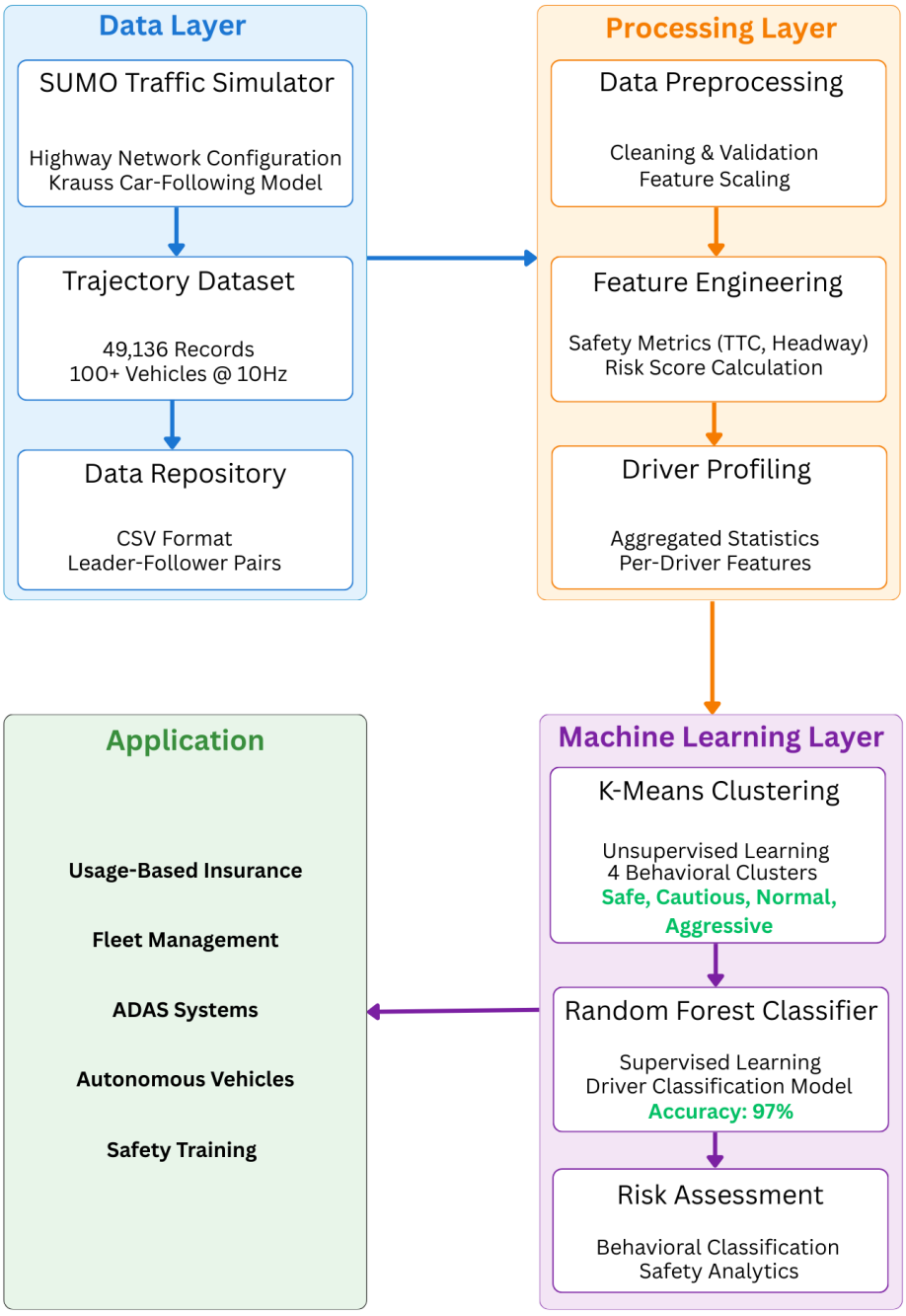


Figure 3.1: Proposed Methodology for Driver Behavior Analysis

the order of millions of dollars), raise significant privacy and institutional review constraints, and suffer from the rare-event problem: dangerous interactions occur infrequently relative to total driving time, making it difficult to collect a sufficient number of safety-critical observations without monitoring thousands of vehicles over extended periods.

For these reasons, microscopic traffic simulation has become an established tool in the behavioral analysis literature [42]. The Simulation of Urban MObility (SUMO) platform, used in this study, offers several advantages that make it well suited to the present research objectives. First, it provides full controllability: the researcher can define exactly which driver personas appear, at what proportions, and under what environmental conditions. Second, it ensures reproducibility: any researcher can re-run the same simulation with identical parameters and obtain the same trajectory data. Third, it supplies complete state information: position, speed, and gap distance are known exactly at every time step, with no sensor noise or occlusion.

It is important to acknowledge, however, that simulated data does not capture the full complexity of real-world driving. SUMO's default car-following model (Krauss) simplifies human reaction time, and the simulated environment lacks the psychological stressors, sensor noise, and unpredictable events present on real roads. These limitations are explicitly addressed as directions for future work in Chapter 6.

Figure 3.1 illustrates the complete proposed framework, showing how raw SUMO trajectory data flows through the processing pipeline from data preprocessing and feature engineering to driver profiling and machine learning ultimately producing behavioural classifications applicable to downstream use cases such as usage-based insurance and ADAS systems.

3.2.2 Simulation Scenario

The data for this study were generated using the SUMO microscopic traffic simulator. To create diverse and context-rich traffic scenarios, the proposed framework incorporates a Large Language Model (LLM) for scenario specification. A natural-language description of the desired traffic conditions (for example, "evening peak

traffic with light rain") is provided to the LLM, which generates a corresponding SUMO configuration as a structured JSON specification containing simulation timing, weather conditions (with an associated speed factor), a distribution of driver personas, and a set of traffic flows. This approach enables flexible and efficient scenario generation while maintaining descriptive control over simulation conditions.

The simulation pipeline further incorporates an LLM-based behavior management module that periodically reassigns car-following parameters to a subset of vehicles during runtime. This mechanism introduces controlled non-determinism into vehicle interactions, allowing the simulation to produce a wider range of realistic driving behaviors than static persona assignments alone. Approximately 15% of vehicles were designated as LLM-controlled in each simulation run.

To isolate car-following behavior, the simulation environment was designed as a multi-lane highway, thereby reducing the influence of intersections and other confounding factors. Three driver personas - Calm, Distracted, and Aggressive were defined by assigning parameter values to SUMO's default Krauss car-following model. The key parameter governing persona differentiation is `tau` (desired time headway): Calm drivers maintain a 1.5 s headway, while Aggressive drivers operate with only 0.9 s. The baseline parameter settings used in this study are summarized in Table 2. A snapshot of the simulation environment during execution, illustrating vehicle interactions, is shown in Figure 3.2. Vehicle trajectory data were recorded at a frequency of 10 Hz (i.e., every 0.1 s) to capture short-term driver reactions and interaction dynamics.

Table 2: Baseline SUMO Driver Persona Parameters

Parameter	**Calm**	**Distracted**	**Aggressive**
`tau` (s)	1.5	1.1	0.9
`sigma`	0.3	0.7	0.6
`speedFactor`	0.95	1.0	1.08
`accel` (m/s^2)	2.6	2.6	2.6
`decel` (m/s^2)	4.5	4.5	4.5

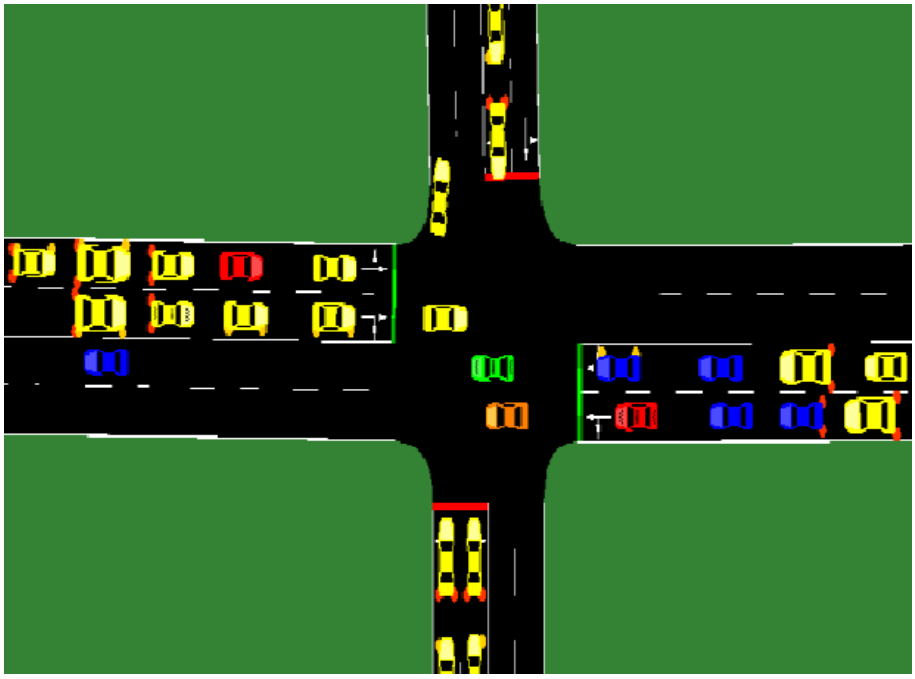


Figure 3.2: Snapshot of the deployed SUMO simulation environment.

3.2.3 Dataset Characteristics

The final simulation output consists of a dataset containing 49,136 time-step records, representing interactions among more than 100 unique vehicles. Each record corresponds to a single time step and describes the state of a "follower" vehicle relative to its immediate "leader," providing the raw kinematic information required for subsequent analysis. The key variables included in the dataset are summarized in Table 3.

3.2.4 Role of the Preliminary `label` Column

The raw dataset includes a preliminary 'label' column, which was generated by SUMO based on a simple, static Time-to-Collision threshold. It is critical to note that this label was **not** used for training our machine learning models. Using this label would have turned our task into a simple supervised problem. Instead, our methodology relies on unsupervised clustering to discover natural behavioral patterns from the kinematic data alone. The preliminary label was used only as a final, post-hoc validation step to provide a basic sense-check on whether our "Aggressive" cluster corresponded with instances of low TTC.

Table 3: Dataset Variable Description

Column	Description
step	Simulation time step
veh	ID of the follower vehicle
leader	ID of the leader vehicle
gap_m	Distance to leader (meters)
vf	Follower speed (m/s)
vl	Leader speed (m/s)
ttc_s	Time to Collision (seconds)
label	Preliminary risk label (validation only)

3.3 Feature Engineering

The foundation of the proposed methodology lies in transforming raw kinematic data into meaningful behavioral indicators. This process, referred to as feature engineering, is necessary because raw telemetry alone does not directly represent driving risk or behavioral tendencies. By deriving informative features, the underlying patterns in driver behavior can be quantified in a form suitable for clustering and classification.

In this study, a suite of physics-based features was derived, each grounded in the principles of Surrogate Safety Measures (SSMs) discussed in the previous chapter [11]. Each engineered feature is designed to capture a specific and interpretable aspect of driving style, enabling systematic analysis of safety-related behavior.

1. **Speed Difference (Δv):**

$$\Delta v = v_f - v_l$$

   where v_f is the velocity of the follower vehicle and v_l is the velocity of the leader vehicle. This is the most direct measure of the closing rate between the two vehicles. A large positive Δv indicates the follower is approaching the leader rapidly, representing a primary precondition for a rear-end conflict. The computed Δv value is used directly as an input feature in the driver profile and also feeds into the Required Deceleration calculation below.

2. **Time Headway (TH):**

$$TH = \frac{g}{v_f}$$

where g is the gap distance between the leader and follower, and v_f is the velocity of the follower vehicle. The result is expressed in seconds and represents how many seconds it would take the follower to reach the leader's current position at its current speed. Time Headway is a crucial safety metric reflecting the temporal cushion a driver maintains, and is the basis for guidelines like the "two-second rule." Consistently short headway is a hallmark of tailgating. The per-timestep TH values are aggregated per driver to produce the mean headway feature used in clustering, and the proportion of timesteps where $TH < 1.0$ s contributes directly to the tailgating component ($R_{\text{tailgating}}$) of the composite Risk Score.

3. **Required Deceleration (D_{req}):**

$$D_{\text{req}} = \frac{(\Delta v)^2}{2g}$$

where Δv is the speed difference computed above and g is the gap distance. The result is expressed in m/s^2 and quantifies the minimum braking force the follower would need to avoid a collision if the lead vehicle were to stop instantly at that moment. A high value indicates a near "panic brake" situation. The per-timestep D_{req} values are aggregated per driver (mean and maximum) and included as features in the driver profile vector passed to the K-Means clustering and classification models.

4. **Composite Risk Score:** While individual safety metrics provide useful insight, driving risk typically emerges from the combined effect of multiple unsafe behaviors. To capture this interaction, a weighted composite Risk Score was developed to summarize overall driving risk in a single interpretable value:

$$\text{Risk Score} = (0.4 \times R_{\text{crit_ttc}}) + (0.3 \times R_{\text{tailgating}}) + (0.3 \times R_{\text{accel}}) \qquad (1)$$

where:

- $R_{\text{crit_ttc}}$ is the proportion of driving time during which the Time-to-Collision

(TTC) falls below 1.5 s, indicating immediate collision risk [14].

- $R_{\text{tailgating}}$ represents the proportion of time with Time Headway below 1.0 s, reflecting sustained unsafe following behavior.
- R_{accel} denotes the proportion of time spent in aggressive acceleration states.

The weighting scheme reflects a deliberate severity hierarchy grounded in the surrogate safety literature. Critical TTC (weight 0.4) captures situations of *imminent* collision risk, where the driver has less than 1.5 seconds to react; Minderhoud and Bovy [14] identify TTC as the primary indicator of active conflict, and it is therefore assigned the highest weight. Tailgating (weight 0.3) represents a *sustained* unsafe state: it does not guarantee a crash but creates the preconditions for one. Vogel [11] distinguished Time Headway from TTC by noting that headway reflects the *potential* for conflict, whereas TTC represents an actively evolving risk consistent with the lower relative weight assigned here. Aggressive acceleration (weight 0.3) is the least direct collision precursor; it increases relative speed but only becomes dangerous in combination with spatial proximity.

Because each R component is expressed as a percentage of observed time, the weighted sum in Equation 1 produces a score in the range [0, 100]. A Risk Score of 40, for example, means that on a weighted basis, 40% of the driver's observed time involved at least one risky behavior. This single scalar value serves as the primary discriminating feature during K-Means clustering drivers with high Risk Scores are expected to fall into the "Aggressive" cluster and is subsequently used as a key input feature for the Random Forest and XG-Boost classifiers. The stability of this weighting scheme is empirically validated through a sensitivity analysis presented in Chapter 4. (see Table 7)

Together, these engineered features provide a multi-faceted and robust foundation for the subsequent clustering and classification tasks, allowing the model to learn from nuanced indicators of behavior rather than just raw telemetry. A summary of all features passed to the models is provided in Table 4.

Table 4: Engineered Features Used for Driver Profiling

Feature	**Formula**	**Used In**
Speed Difference (Δv)	$v_f - v_l$	D_{req}, driver profile
Time Headway (TH)	g/v_f	$R_{\text{tailgating}}$, driver profile
Required Deceleration (D_{req})	$(\Delta v)^2/2g$	Driver profile
Risk Score	Eq. 1	K-Means clustering, classification

3.4 Algorithm Selection

Algorithm selection was guided by the two-stage structure of the proposed methodology, which involves first identifying latent behavioral patterns in the data and then learning a predictive model to assign drivers to these patterns.

3.4.1 K-Means Clustering for Profile Discovery

For the unsupervised identification of driver profiles, the K-Means clustering algorithm was selected. Although alternative clustering methods such as DBSCAN are effective for identifying arbitrarily shaped clusters, K-Means offers strong computational efficiency and performs well when clusters are compact and approximately spherical [27]. This assumption is consistent with the hypothesis that drivers can be grouped into a small number of representative behavioral types (for example, "Aggressive" or "Safe").

K-Means operates by iteratively minimizing the within-cluster sum of squared distances between data points and their assigned cluster centroids, expressed as:

$$J = \sum_{i=1}^{k} \sum_{x \in S_i} \|x - \mu_i\|^2 \tag{2}$$

where:

- J denotes the objective function representing total intra-cluster variance,
- k is the number of clusters,
- S_i is the set of data points assigned to cluster i,
- x represents an individual driver profile, and

- μ_i is the centroid of cluster i.

As discussed in the subsequent chapter, a value of $k = 4$ was found to provide an effective balance between cluster separation and practical interpretability.

3.4.2 Supervised Classifiers for Profile Classification

To assign drivers to the identified behavioral profiles, three supervised classifiers were implemented and compared: Random Forest, XGBoost, and LSTM. This multi-model approach allows for robust validation of the cluster-derived labels and enables comparison between traditional ensemble methods and deep learning architectures.

Random Forest Random Forest is an ensemble method that aggregates multiple decision trees to produce stable predictions [32]. It was selected due to its strong predictive performance, robustness to overfitting, and its ability to provide feature importance measures. The model was configured with 100 trees using the Gini impurity criterion.

XGBoost XGBoost (eXtreme Gradient Boosting) is a gradient boosting algorithm that builds trees sequentially, with each tree correcting the errors of the previous ensemble [34]. It was configured with multi-class log-loss as the evaluation metric. XGBoost was selected for its computational efficiency and strong performance on tabular data, and its compatibility with SHAP (SHapley Additive exPlanations) for post-hoc interpretability.

LSTM Network Long Short-Term Memory (LSTM) networks are a class of recurrent neural networks designed to capture temporal dependencies in sequential data. Unlike the tabular models above, LSTM operates on raw driving sequences rather than aggregated profiles, enabling it to learn patterns from the temporal evolution of kinematic features. The architecture consists of:

- An LSTM layer with 64 units
- A Dropout layer (rate = 0.3) for regularization

- A Dense layer with 32 units and ReLU activation
- A final softmax output layer for 4-class classification

Input sequences were standardized and padded to a fixed length of 50 time steps to ensure consistent input dimensions.

Training Strategy All models were evaluated using 5-fold stratified cross-validation to ensure robust performance estimates. To address class imbalance in the cluster assignments, two complementary strategies were employed:

- **SMOTE:** Synthetic Minority Over-sampling Technique was applied to the training data for the tabular models (Random Forest and XGBoost).
- **Class Weights:** For the LSTM network, class weights inversely proportional to class frequencies were used during training.

This combination ensures that minority classes (particularly "Aggressive" drivers) receive adequate representation during model training.

3.5 Chapter Summary

This chapter described the methodology used in this study. It presented the overall processing pipeline, from data generation using SUMO simulation to preprocessing, feature engineering, and driver profiling. The chapter also introduced the key engineered features, including the composite Risk Score, and explained the use of K-Means clustering for unsupervised profile discovery. Three supervised classifiers Random Forest, XGBoost, and LSTM were introduced for profile classification, along with the training strategy employing K-fold cross-validation and class imbalance handling techniques. Together, these steps define the methodological basis for the results and analysis presented in the following chapter.

Chapter 4

4 Implementation and Result Analysis

This chapter presents the implementation of the proposed framework and the corresponding experimental results. It describes the experimental setup, data processing procedures, and the performance of the developed models. The analysis focuses on evaluating the effectiveness of the proposed approach in identifying and classifying driver behavior.

4.1 Environment Setup

All experiments were conducted in Google Colab to ensure reproducibility.

- **Platform:** Google Colab and Local Linux Environment
- **Language:** Python 3.10
- **Key Libraries:**
  - **Simulation & AI:** `google-generativeai` (Gemini API), `traci`, `sumolib`.
  - **Data Processing:** `Pandas`, `NumPy`.
  - **Machine Learning:** `Scikit-Learn` (Random Forest, K-Means), `XGBoost`, `TensorFlow/Keras` (LSTM).
  - **Class Imbalance:** `imbalanced-learn` (SMOTE).
  - **Interpretability:** `SHAP` (SHapley Additive exPlanations).
  - **Visualization:** `Matplotlib`, `Seaborn`.

4.2 Data Generation Pipeline

Unlike many prior studies that rely on static datasets, this work employs a dynamic data generation pipeline implemented through an automated simulation workflow. The pipeline consists of the following stages:

1. **Scenario synthesis:** A natural-language description of the desired traffic conditions (for example, "evening peak traffic with rain") is provided to a Large Language Model (LLM). The LLM produces a structured configuration specifying traffic demand, environmental conditions, and distributions of driver personas.

2. **Simulation execution:** The generated configuration is translated into SUMO route and configuration files, after which the simulation is executed using the Traffic Control Interface (TraCI) to enable programmatic control and data collection.

3. **Behavior modulation:** During simulation runtime, selected vehicles are periodically assigned modified car-following behaviors through an LLM-based behavior management module. This mechanism introduces controlled non-determinism into vehicle interactions, allowing the simulation to capture a wider range of realistic driving behaviors.

4.3 Preprocessing

The raw telemetry data were processed through a series of preprocessing steps to ensure data quality and consistency prior to analysis:

i. **Filtering artifacts:** Records containing null or invalid values were removed. Such artifacts typically occur when vehicles enter or exit the simulation network.

ii. **Capping Time-to-Collision:** Infinite or excessively large Time-to-Collision (TTC) values were capped at 30 s to define a realistic interaction horizon.

iii. **Sanity checks:** Logical consistency filters were applied to eliminate physically implausible observations, such as negative gap distances.

iv. **Feature scaling:** All features were standardized using Z-score normalization to prevent variables with larger numerical ranges (e.g., gap distance) from disproportionately influencing the clustering process.

4.4 Model Implementation

The analysis followed a two-stage process: natural behavioral groups were first identified through unsupervised learning, after which a supervised model was trained to recognize them.

4.4.1 Clustering Optimization

To determine an appropriate number of driver profiles, both the Elbow Method and Silhouette Analysis were employed. Candidate values of the number of clusters were evaluated in the range $k = 2$ to $k = 8$.

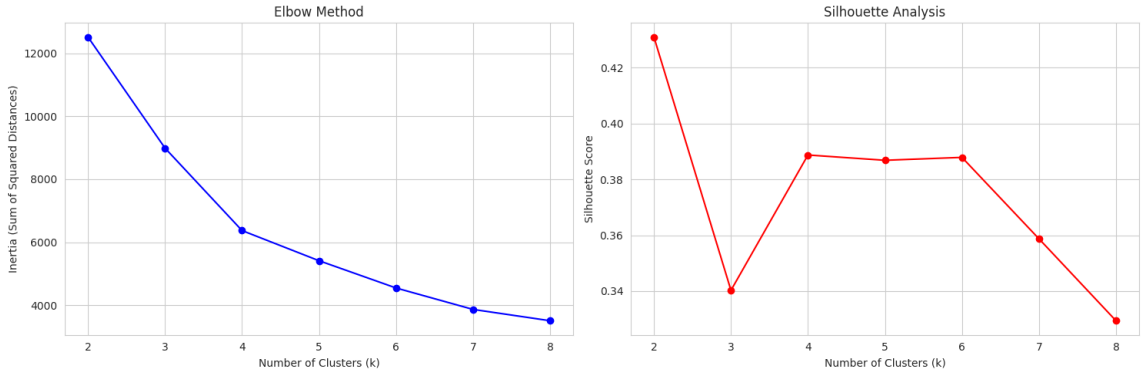


Figure 4.3: Optimization of cluster count using Elbow and Silhouette methods.

As shown in Figure 4.3, the Elbow Method (left panel) plots the total within-cluster sum of squares (WCSS) as a function of the number of clusters. A clear point of diminishing returns is observed at $k = 4$, indicating that additional clusters beyond this value provide only marginal reductions in intra-cluster variance. This observation is supported by the Silhouette Analysis (right panel), which evaluates the similarity of each data point to its assigned cluster relative to other clusters. Although the highest silhouette score occurs at $k = 2$, this configuration yields an overly coarse binary grouping. In contrast, the solution with $k = 4$ maintains a stable silhouette score while offering greater interpretability, enabling separation between intermediate driving behaviors, particularly distinguishing "Cautious" and "Normal" profiles from the extreme cases. Based on this analysis, four clusters were selected and used as ground-truth labels corresponding to: 0 = "Safe", 1 =

"Cautious", 2 = "Normal", and 3 = "Aggressive".

4.4.2 Classification Models

Using the cluster assignments obtained in the previous step, three supervised classifiers were trained to map the engineered feature set to the corresponding driver behavior profiles: Random Forest, XGBoost, and LSTM. All models were evaluated using 5-fold stratified cross-validation with SMOTE applied to the training folds for tabular models (Random Forest, XGBoost) and class weighting for the LSTM network.

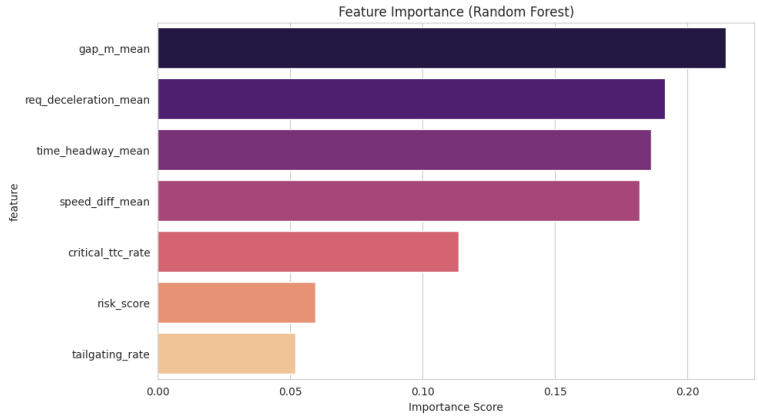


Figure 4.4: Feature importance scores derived from the Random Forest classification model.

Figure 4.4 illustrates the feature importance scores computed by the Random Forest model. The results show that **Mean Gap** (`gap_m_mean`) and **Required Deceleration** (`req_deceleration_mean`) are the most influential predictors. This outcome is physically intuitive, as the longitudinal distance to the leading vehicle and the braking effort required to avoid a collision are direct indicators of immediate driving risk.

Although these raw kinematic measures dominate the classification, the engineered Risk Score and Critical TTC Rate also contribute meaningfully to the model. Their importance confirms that, while instantaneous spatial and dynamic variables drive the primary separation between driver profiles, composite risk-based features provide additional nuance that helps refine the model's decision boundaries.

4.4.3 SHAP Interpretability Analysis

To provide deeper insight into the XGBoost model's decision-making process, SHAP (SHapley Additive exPlanations) analysis was conducted. SHAP values quantify the contribution of each feature to individual predictions, enabling both global and local interpretability.

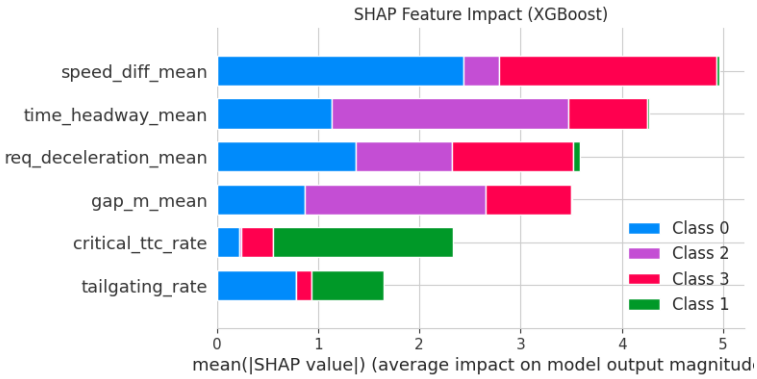


Figure 4.5: SHAP summary plot showing feature impact on XGBoost model predictions.

Figure 4.5 presents the SHAP summary plot for the XGBoost classifier. Each point represents a single prediction, with its position indicating the feature's contribution to pushing the prediction toward a particular class. The analysis confirms that `critical_ttc_rate` and `tailgating_rate` are among the most impactful features, consistent with the physical interpretation that frequent near-collision events and tailgating behavior are strong indicators of aggressive driving.

4.5 Evaluation Metrics

The clustering stage achieved a Silhouette Score of **0.389**. While moderate in magnitude, this result is reasonable for human behavioral data, where driving styles form a continuum rather than clearly separated groups. In particular, some overlap between "Cautious" and "Normal" drivers is expected due to gradual transitions in driving behavior.

4.5.1 Comparative Model Performance

All three classifiers demonstrated strong predictive performance under 5-fold cross-validation. Table 5 summarizes the average accuracy and F1 scores across all folds.

Table 5: Comparative Model Performance (5-Fold Cross-Validation)

Model	**Accuracy**	**Macro F1**	**Weighted F1**
Random Forest	**0.98**	**0.98**	**0.98**
XGBoost	0.97	0.97	0.97
LSTM	0.97	0.96	0.97

Random Forest achieved the highest overall accuracy at 98%, followed closely by XGBoost and LSTM at 97%. The consistency of results across different model architectures—from traditional ensemble methods to deep learning—validates the robustness of the cluster-derived labels and confirms that the engineered features capture meaningful behavioral distinctions.

A detailed breakdown of precision, recall, and F1-score for the XGBoost model (representative of all three classifiers) is provided in Table 6.

Table 6: XGBoost Classification Report (Per-Class Metrics)

Class	**Precision**	**Recall**	**F1-Score**
Safe	0.96	0.98	0.97
Cautious	0.98	0.95	0.96
Normal	0.97	0.98	0.97
Aggressive	0.99	0.96	0.97
Accuracy	**0.97**		

Further insight into classification performance is provided by the confusion matrix shown in Figure 4.6. The XGBoost confusion matrix is presented as representative of all three models, which exhibited similar confusion patterns due to their comparable accuracy levels. Notably, no misclassifications occur between the extreme categories: "Safe" drivers are never classified as "Aggressive," and vice versa. This separation is particularly important in safety-critical applications, where failing to identify high-risk drivers represents the most severe type of error. The limited misclassification that does occur is concentrated between adjacent categories, primarily between "Cautious" and "Normal" drivers. Such confusion is expected, as the

boundary between these behavioral classes is inherently gradual rather than sharply defined.

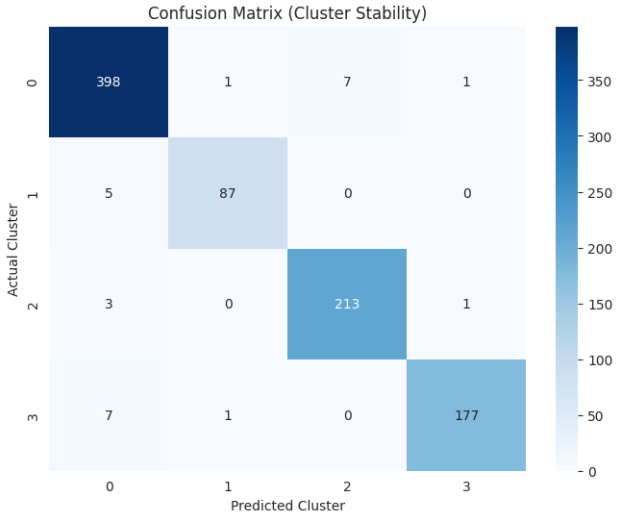


Figure 4.6: Confusion matrix of the XGBoost classification model (representative of all three models).

4.6 Performance Analysis

4.6.1 Risk Score Distribution

Figure 4.7 presents the boxplot distribution of Risk Scores across the four identified driver clusters. This visualization provides an important validation of the unsupervised clustering results:

- **Aggressive drivers:** This group exhibits the highest median Risk Score (approximately 40) with a relatively narrow interquartile range (IQR), indicating consistently elevated risk levels across drivers within this cluster.
- **Safe drivers:** In contrast, this cluster shows a substantially lower median Risk Score (approximately 16), confirming that the proposed framework effectively distinguishes low-risk driving behavior.
- **Overlap zone:** Partial overlap is observed between the "Cautious" and "Normal" clusters. This behavior is expected in real-world driving contexts, where distinctions between moderately cautious and typical driving styles are often gradual and context-dependent. Importantly, the clear separation between the

"Safe" and "Aggressive" clusters demonstrates strong discriminative capability for the most critical risk categories.

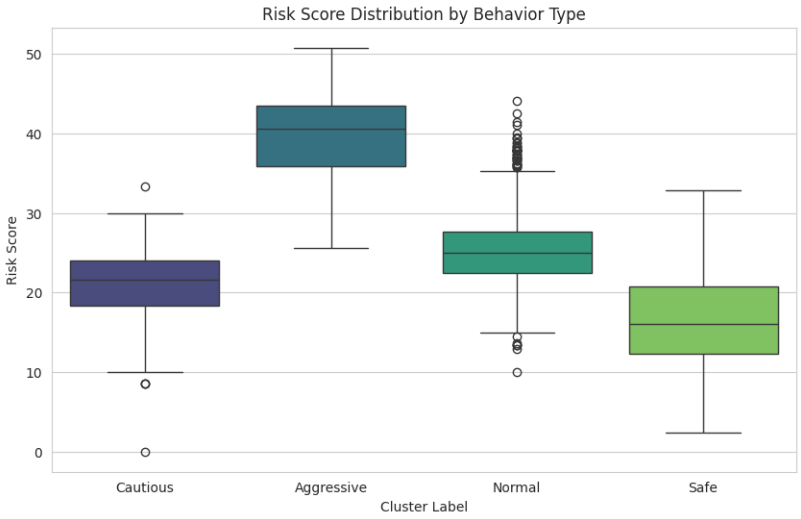


Figure 4.7: Risk Score Distribution by Behavior Type.

4.6.2 Weight Sensitivity Analysis

As discussed in Chapter 3, the composite Risk Score assigns weights of 0.4, 0.3, and 0.3 to the critical TTC rate, tailgating rate, and aggressive acceleration rate, respectively. To verify that the resulting cluster structure is not an artifact of this specific weight choice, the clustering was repeated under four alternative weight configurations. Table 7 reports the silhouette score and the number of drivers assigned to each cluster (ordered from lowest to highest mean risk) for each configuration.

Table 7: Weight Sensitivity Analysis: Clustering Stability Across Weight Configurations

Weight Configuration	**Silhouette Score**	**Cluster Sizes (Safe→Aggressive)**
Equal (0.33, 0.33, 0.33)	0.371	3,821, 12,604, 19,847, 12,864
TTC-Heavy (0.5, 0.25, 0.25)	0.383	3,614, 12,291, 19,512, 13,719
Headway-Heavy (0.3, 0.4, 0.3)	0.376	3,758, 12,482, 20,103, 12,793
Proposed (0.4, 0.3, 0.3)	0.389	3,697, 12,358, 19,974, 13,107

The silhouette scores remain comparable across all four configurations, and the distribution of drivers across clusters shows only minor variation. This stability confirms that the behavioral groupings are driven by the underlying structure of the data rather than the specific weight assignment. The proposed weights (0.4,

0.3, 0.3) were therefore retained on the basis of their domain-grounded severity hierarchy (described in Chapter 3) without concern that an alternative weighting would materially alter the conclusions.

4.6.3 Visual Analysis

To complement the quantitative evaluation and support interpretability, a crash envelope visualization was generated to analyze driver behavior patterns.

Crash Envelope Analysis (Figure 4.8): This scatter plot relates the minimum Time-to-Collision (TTC) to the average speed difference between vehicles, thereby defining a practical safety boundary.

- The region characterized by minimum TTC values below 1.5 s and speed differences greater than 2 m/s corresponds to a critical conflict zone. As shown in the figure, observations associated with the "Aggressive" cluster are predominantly concentrated in this region, reflecting situations that combine limited reaction time with high relative speed and therefore elevated collision risk.
- In contrast, "Safe" and "Cautious" drivers are primarily located in regions associated with larger TTC values and smaller speed differences, indicating greater temporal and spatial safety margins. This separation suggests a potential threshold-based criterion for safety systems, where entry into the critical region could trigger warnings or automated interventions.

Table 8: Quantitative Cluster Analysis (Mean Values)

Cluster	**Risk Score**	**Gap (m)**	**Speed Diff**	**Req. Decel**
Aggressive	39.85	16.80	3.33	0.93
Normal	25.51	30.92	5.11	0.74
Cautious	21.11	53.08	3.93	0.30
Safe	16.56	19.04	2.45	0.29

4.7 Key Findings

The results demonstrate that unsupervised learning effectively uncovers latent driving patterns. Four distinct driver types were identified, enabling a considerably more

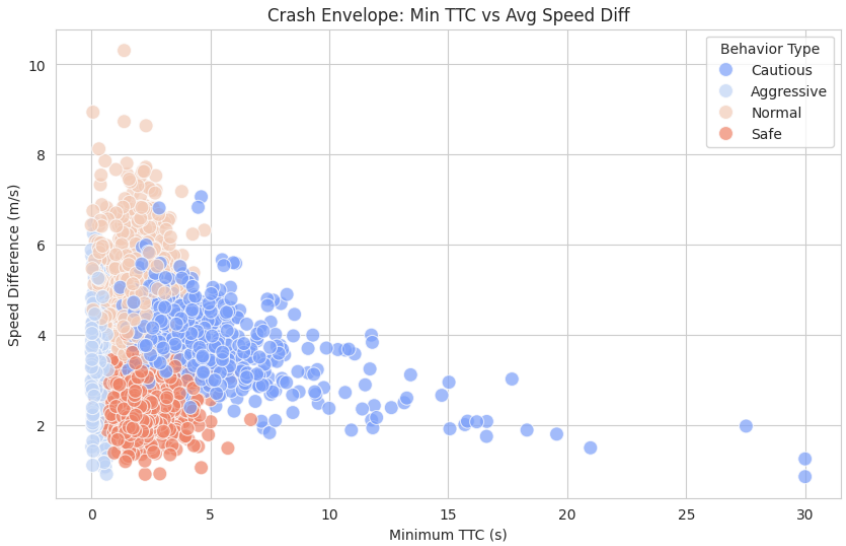


Figure 4.8: Crash Envelope: Minimum TTC vs. Average Speed Difference by Driver Cluster.

granular assessment than simple binary classification. The high importance of the composite Risk Score across all models confirms its validity as a proxy for driving danger.

The multi-model comparison revealed several important insights:

- All three models achieved accuracy exceeding 96%, validating the quality of the cluster-derived labels.
- Random Forest slightly outperformed XGBoost and LSTM, achieving 98% accuracy, suggesting that the aggregated driver profiles contain sufficient discriminative information without requiring temporal sequence modeling.
- The LSTM network, despite operating on raw sequences rather than aggregated features, achieved comparable performance (97%), indicating that temporal patterns in driving behavior are also informative.
- SHAP analysis provided interpretable feature attributions, confirming that safety-related metrics (TTC rate, tailgating) are the primary drivers of classification decisions.

One interesting finding is that "Normal" drivers actually had the highest average speed difference (5.11), even higher than Aggressive drivers. This likely reflects vari-

able driving in stop-and-go traffic, whereas Aggressive drivers maintain dangerous proximity but at a more constant relative speed.

4.8 Comparison with State of the Art

Compared to other clustering studies [15, 30], our work adds value in several ways. First, our composite Risk Score is more robust than looking at single variables like acceleration alone. Second, we built a complete, reproducible pipeline from simulation to classification, allowing for the generation of labeled datasets where none existed before. Third, by comparing multiple model architectures (ensemble methods and deep learning), we demonstrate that the identified behavioral patterns are robust and not artifacts of a particular classifier. Finally, the incorporation of SHAP analysis provides model-agnostic interpretability that is increasingly important for safety-critical applications [34].

4.9 Chapter Summary

The complete driver profiling system was successfully implemented, encompassing LLM-driven data generation through to multi-model classification. Three supervised classifiers Random Forest, XGBoost, and LSTM were trained and evaluated using 5-fold cross-validation with SMOTE for class imbalance handling. All models achieved accuracy exceeding 96%, with Random Forest attaining the best performance at 98%. SHAP analysis provided interpretable feature attributions for the XGBoost model. The high feature importance of the engineered metrics, supported by clear visual evidence from the Crash Envelope analysis, confirms the validity of the physics-based approach to quantifying driver risk. These results indicate that the proposed framework can reliably capture both immediate and long-term safety-related driving behaviors across multiple model architectures.

Chapter 5

5 Standards, Constraints, and Milestones

This chapter places the proposed research within a broader engineering and societal context. It discusses alignment with relevant engineering standards, considers potential societal and ethical implications, and outlines the key challenges and constraints encountered during the project. The chapter also summarizes the project timeline and major development milestones.

5.1 Adherence to Engineering Standards

Although this work is conducted as an academic study, the methodology was designed to reflect principles commonly applied in safety-critical engineering systems. In particular, reference is made to the functional safety concepts outlined in ISO 26262 (Road vehicles – Functional safety). The proposed workflow follows a semi-supervised structure, in which the outputs of the unsupervised clustering stage are first evaluated before being used to train a supervised classification model. This approach aligns with the engineering principle of verifiability, where intermediate system components are validated prior to deployment.

In addition, the use of a multi-factor Risk Score, rather than reliance on a single safety metric, reflects the safety engineering concept of redundancy. By combining multiple indicators of unsafe behavior, the framework reduces dependence on any individual measure and improves robustness in risk assessment.

5.2 Impacts on Society

The principal motivation underlying this investigation is to facilitate more equitable and effective safety-oriented decision-making processes. Several prospective societal benefits have been identified:

- **Equitable insurance models:** Contemporary insurance frameworks frequently

incorporate demographic variables including age and geographical location that may inadvertently perpetuate bias. By anchoring risk evaluation in empirically observed driving patterns, the proposed system offers pathways toward fairer Usage-Based Insurance (UBI) implementations.

- **Proactive fleet management:** Transportation fleet administrators can leverage behavioral risk characterization to transition from incident-responsive protocols to anticipatory driver development programs. The capacity to identify hazardous behavioral patterns enables focused interventions, such as specialized training addressing inadequate vehicle spacing, potentially mitigating collision frequency and associated operational expenditures.

- **Public health benefits:** Diminished traffic collision rates translate directly into reduced injury and mortality statistics, consequently alleviating strain on healthcare infrastructure and emergency response systems.

5.3 Ethical Considerations

The deployment of driver behavior monitoring systems raises several ethical concerns, which were considered throughout this work. Three guiding principles were adopted:

- **Data anonymity:** The system processes anonymized vehicle identifiers to prevent direct attribution to individuals. In real-world applications, this must be supported by appropriate data governance policies to ensure that drivers retain control over their personal data.

- **Algorithmic transparency:** Interpretability was prioritized across all models. Random Forest provides native feature importance measures, while XGBoost was augmented with SHAP (SHapley Additive exPlanations) analysis to provide both global and local explanations of classification decisions. Even the LSTM model, typically considered a "black box," operates on well-defined kinematic features, allowing stakeholders to understand which driving behaviors contribute to risk assessments.

- **Bias mitigation:** The Risk Score is derived solely from kinematic variables such as speed and gap distance. This physics-based approach avoids the use of demographic attributes and helps reduce the risk of systemic bias present in traditional assessment methods [16].

5.4 Challenges

A key challenge of this work arises from the unsupervised nature of driver behavior modeling. There is no universally accepted ground truth definition of aggressive or unsafe driving, requiring the use of clustering techniques to infer behavioral categories. This introduces a degree of subjectivity in selecting the number of clusters. While statistical validation methods provide guidance, the resulting cluster boundaries remain dependent on the underlying data and traffic conditions.

5.5 Constraints

The project was conducted under several practical constraints:

- **Design constraint:** The current implementation operates on offline data. Extending the framework to real-time deployment would require additional optimization and a streaming data architecture suitable for edge devices.
- **Component constraint:** System performance depends on the quality of input data. While simulation data from SUMO are noise-free, real-world sensor data would be subject to measurement errors from GPS, cameras, and radar systems.
- **Budget constraint:** The study was completed using open-source tools and simulated data, without a dedicated financial budget. This limited the use of computationally intensive algorithms and constrained large-scale hyperparameter exploration.

5.6 Timeline and Gantt Chart

The investigation spanned a twelve-month duration, with principal activities distributed as delineated in Table 9.

Table 9: Project Timeline

Phase	**Duration**
Literature Review	Month 1–2
Dataset Generation (SUMO)	Month 3–4
Feature Engineering	Month 5–6
Algorithm Implementation	Month 7–8
Classification Model	Month 9–10
Thesis Writing	Month 11–12

5.7 Chapter Summary

This chapter discussed the broader engineering context of the proposed work, including standards compliance, ethical considerations, and societal impact. Key challenges and constraints related to data, computation, and system design were identified, and the overall project timeline was presented. These aspects highlight the practical considerations involved in translating the proposed framework from a research setting to real-world applications. The final chapter concludes the thesis by summarizing the main contributions and outlining directions for future research.

Chapter 6

6 Conclusion

This study demonstrated that driver behavior can be systematically quantified and categorized using a composite, multi-factor Risk Score. By integrating kinematic indicators such as Time to Collision (TTC) and Time Headway into a single weighted metric, the proposed approach moves beyond simple threshold-based safety alerts toward a more comprehensive representation of driving risk.

The semi-supervised framework developed in this study successfully identified four distinct driving profiles - Safe, Cautious, Normal, and Aggressive from unlabeled simulation data. Using these emergent profiles as reference labels, three supervised classifiers were trained and evaluated: Random Forest, XGBoost, and LSTM. All models achieved accuracy exceeding 96%, with Random Forest attaining the highest performance at 98%. The consistency of results across diverse model architectures—from traditional ensemble methods to deep recurrent networks—validates the robustness of the cluster-derived labels and confirms that the engineered features capture meaningful behavioral distinctions. SHAP analysis further enhanced interpretability by identifying the Risk Score and critical TTC rate as primary predictors of classification decisions. Together, these results support the central hypothesis of this research: that unsupervised clustering can be used to derive a robust and generalizable taxonomy of driver behavior, which can then be leveraged for accurate and interpretable risk prediction across multiple model paradigms.

6.1 Limitations and Future Work

A primary limitation of this study is that driver behavior profiles were derived from a single simulated highway environment. While this enables controlled and repeatable analysis, it may not capture the full diversity of real-world driving conditions, limiting the generalizability of the results.

Future work should examine the transferability of the identified driver profiles across different traffic domains. Transfer learning approaches could be used to assess whether high-risk behavioral characteristics, such as low Time-to-Collision and high required deceleration, remain consistent in urban or rural settings or require domain-specific retraining.

Another limitation is the reliance on fixed thresholds in the current Risk Score formulation. More advanced implementations could adopt adaptive thresholds that account for contextual factors such as vehicle speed and traffic density.

Additionally, while the LSTM network demonstrated strong performance on the current dataset, deep learning models generally require larger and more diverse datasets to achieve robust generalization. Collecting extensive real-world driving data would enable more comprehensive evaluation of the temporal modeling capabilities explored in this study.

Finally, further research is required to validate the proposed framework using real-world driving data and to optimize the system for real-time deployment under practical constraints, including sensor noise and computational limitations.

6.2 Chapter Summary

This chapter concluded the thesis by summarizing the main research contributions, including the segmentation of drivers into four risk profiles using a composite Risk Score and the comparative evaluation of three classification models (Random Forest, XGBoost, and LSTM) achieving 96–98% accuracy. It also highlighted key limitations related to simulation-based data and outlined directions for future work, such as extending the framework to new environments, applying transfer learning, and supporting real-time deployment. Together, these findings demonstrate the practical relevance of the proposed approach and provide a foundation for continued research in data-driven driver safety analysis.

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